Chenchu Yakasiri Saravanan

US Permanent Resident | ynchenchuh@icloud.com | (310) 357-4896 | Website: chenchu.dev

EDUCATION

California State University, Long Beach — Long Beach, California
B.S. Physics — *Expected Graduation: Dec 2027*
Concentration: Experimental Condensed Matter Physics

EXPERIENCE

CSULB Nanoelectronics Group — Long Beach, California
Undergraduate Researcher — *Jun 2024 – Present*

- Adapted a hemispherical PC/PDMS dome stamp for localized, high-precision dry transfer of $PtTe_2$ crystals; successfully demonstrated targeted crystal pickup.
- Refined stamp fabrication and transfer parameters, including dome geometry, temperature, contact time, and retraction rate, to improve pickup consistency and target smaller flakes.
- Began adapting the method to $Cr_xPt_{1-x}Te_2$ and developing bottom-contact Hall bars for future electronic transport measurements.
- Authored CSULB's updated Raman spectroscopy SOP and served as a primary Raman instrument operator.

CSULB Shark-Sat 1 Project — Long Beach, California
ADCS Engineer — *Aug 2023 – Nov 2024*

- Led the initial ADCS design draft, working with vendors and industry veterans on orbital dynamics and satellite tracking.
- Contributed to the ADCS team's detumble simulation design to support Avionics software development.

PROJECTS

WeatherCLI — *Mar 2026 – Present*

- Built and maintain a pipx-installable Python weather CLI using Open-Meteo, Geopy, and Rich, with location search, configurable output, unit conversion, automated tests, and GitHub Actions CI.

Raman Spectra Visualization Tool — *May 2025*

- Created Jupyter Notebook tools using NumPy and Plotly to visualize individual and stacked $PtTe_2$ Raman spectra and compare peak regions across samples.

5 mW Laser Pointer PCB — *Oct 2026 – Present*

- Designing a compact, hold-to-activate laser pointer in KiCad, including the schematic and PCB for a AAA-powered circuit that steps up to 3.3 V for a 5 mW red laser module.

KSP Orbital Relay Calculators — *2025*

- Wrote Python calculations for orbital period, eclipse duration, power requirements, and satellite spacing to help plan a Kerbal Space Program relay network.

PRESENTATIONS

APS Global Summit — Poster Presenter — *2025*
APS Far West — Poster Presenter — *2026*
APS Global Summit — Scheduled Oral Presenter — *Apr 2027*

ACTIVITIES AND LEADERSHIP

Western Museum of Flight — Torrance, California
Aircraft Model Restoration Technician and Museum Docent — *Feb 2024 – Jul 2025*
CSULB Astronomy Club — Long Beach, California
Treasurer — *Aug 2024 – Sep 2025*

SKILLS

Programming: Python, Bash, TypeScript/JavaScript, HTML/CSS, MATLAB, LaTeX
Web and Development: Next.js, Cloudflare, AWS, Git/GitHub, website deployment, Jupyter Notebooks
Hardware and Fabrication: KiCad PCB design, 3D printing, SolidWorks
Research and Instrumentation: Python-based cryostat control, Raman spectroscopy, Atomic Force Microscopy, Energy-dispersive X-ray spectroscopy (EDS), WiTec suite, AGI Toolkit